

Visualizing Housing Market Trends: An Analysis of Housing Sale Prices and Property Features Using Tableau

TEAM ID	SWTID -2026-5976
PROJECT TITLE	Visualizing housing market trends: an analysis of sale prices and features using tableau
MAX MARKS	5 MARKS

4.3 – Solution Architecture

data. It aims to provide clear insights into property prices and market trends. The architecture has five main components. This layer collects housing-related data from different sources like real estate websites, property transaction records, government housing datasets, and CSV/Excel files. The data includes property sale prices, The solution architecture is built to collect, process, analyze, and visualize housing market location, square footage, number of bedrooms and bathrooms, year built, lot size, and other property details. The collected data is cleaned and transformed before analysis. It removes or corrects missing values, duplicate records, and inconsistent formats. You can use spreadsheet tools, SQL queries, or Tableau Prep for data preparation to make sure the dataset is accurate and reliable.

After preprocessing, the cleaned dataset is stored in a centralized repository, like a relational database (MySQL, PostgreSQL, SQL Server) or a data warehouse. This layer ensures efficient data management and allows for quick access for analytical processing. The processed data connects to Tableau, where calculations and aggregations happen. Tableau creates interactive visualizations, including line charts, bar charts, scatter plots, heat maps, and geographical maps. These visualizations help identify housing price trends, regional variations, and relationships between property features and sale prices.

The final layer provides dashboards and reports to end users, such as real estate agents, investors, policymakers, and homebuyers. Users can interact with dashboards using filters, drill-down options, and KPI indicators to gain useful insights. The visual reports support decisions about property valuation, investment opportunities, market forecasting, and housing policy planning.

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